Lakshya JEE 2.0 2025

Maths

Relations and Functions

DPP: 11

Q1 Which of the following is an even function
(A) $x\left(\frac{a^x-1}{a^x+1}\right)$
(B) $\tan x$
(C) $\frac{a^x-a^{-x}}{2}$
(D) $\frac{a^x+1}{a^x-1}$

Q2 The function $f(x)=|\sin x|$ has an inverse if its domain is
(A) $[0,\pi]$
(B) $\left[0,\frac{\pi}{2}\right]$
(C) $\left[-\frac{\pi}{4},\frac{\pi}{4}\right]$
(D) $[0,2\pi]$

Q3 If $f(x), f: R \to R$ is an odd periodic function with period 2 , then $f(4)$ equals
(A) 0
(B) 2
(C) 4
(D) -4

Q4 Let $f(x)=\sqrt{x^4+15}$, then the graph of the function $y=f(x)$ is symmetrical about
(A) The x-axis
(B) The y-axis
(C) The origin
(D) The line $x=y$

Q5 Which of the following is an even function
(A) $f(x)=\frac{a^x+1}{a^x-1}$
(B) $f(x)=x\left(\frac{a^x-1}{a^x+1}\right)$
(C) $f(x)=\sin^3 x$
(D) $f(x)=\sin x$

Q6 The function $f(x)=\log\left(x+\sqrt{x^2+1}\right)$ is:
(A) An even function
(B) An odd function
(C) Periodic function
(D) None of these

Q7 Which of the following is not an odd function?
(A) $g(x)-g(-x)$
(B) $(g(x)-g(-x))^3$
(C) $\log\left(\frac{x^4+x^2+1}{x^2+x+1}\right)$
(D) $xg(x)\cdot g(-x)+\tan(\sin x)$

Q8 The period of $f(x)=1+2x-[2x]$, if it is periodic, is
(A) 2
(B) $\frac{1}{2}$
(C) 1
(D) None of these

Q9 Select the correct option
(A) $f(x)=e^{3-2x}\cdot\tan 2x+|x|-\tan x$ is an even function
(B) $f(x)=h(x)+h(-x)+x^2\cot x$ is an even function
(C) $f(x)=\begin{cases} x, & x\in Q \\ -x, & x\notin Q\end{cases}$ is an odd function
(D) $f(x)=g(x)+g(-x)+|\operatorname{sgn}(x)|$ is neither even nor odd function

Q10 The period of the function $f(x)=\sin^4 x+\cos^4 x$ is
(A) π
(B) $\frac{\pi}{2}$
(C) 2π
(D) None of these

Q11 The period of the function $f(x)=\sin x+\tan\frac{x}{2}+\cos\frac{x}{3}$ is

(A) 2π (B) 4π
(C) 6π (D) 8π

Q12 Which of the following function is a periodic function?
(A) $f(x) = \sin\{x\}; \{.\} =$ fractional part of x
(B) $f(x) = \sin[x]; [.] =$ greatest integer function
(C) $f(x) = \log[x]; [.] =$ greatest integer function
(D) $f(x) = \frac{\sin x}{x}; x \neq 0$

Q13 The period of the function $f(x) = |\sin x| + |\cos x|$ is
(A) π
(B) $\pi/2$
(C) 2π
(D) None of these

Q14 The period of the function $f(x) = \sin\left(\frac{2x+3}{6\pi}\right)$, is
(A) 2π
(B) 6π
(C) $6\pi^2$
(D) None of these

Q15 The period of $f(x) = |\sin 2x|$ is:
(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) π
(D) 2π

Q16 The period of the function $f(x) = 2\sin^2 x + 1$ is:
(A) $\frac{\pi}{2}$
(B) π
(C) 2π
(D) None of these

Q17 The function $f(x) = \sin\frac{\pi x}{2} + 2\cos\frac{\pi x}{3} - \tan\frac{\pi x}{4}$ is periodic with period
(A) 6 (B) 3
(C) 4 (D) 12

Q18 Periodic of $f(x) = \sin 3\pi\{x\} + \tan\pi[x]$, where [.] and $\{.\}$denote the greatest integer function and fractional part respectively, is:
(A) 1 (B) 2
(C) 3 (D) None of these

Q19 The period of $f(x) = \sin\left(\frac{\pi x}{n-1}\right) + \cos\left(\frac{\pi x}{n}\right), n \in Z, n > 2$ is:
(A) $2n(n+1)$
(B) $4n(n-1)$
(C) $2n(n-1)$
(D) None of these

Q20 If $f(x) = 2\tan 3x + 5\sqrt{1-\cos 6x}; g(x)$ is a function having the same fundamental period as that of $f(x)$ then which of the following can be $g(x)$.
(A) $\left(\sec^2 3x + \operatorname{cosec}^2 3x\right)\tan^2 3x$
(B) $2\sin 3x + 3\cos 3x$
(C) $2\sqrt{1-\cos^2 3x} + \operatorname{cosec} 3x$
(D) $3\operatorname{cosec} 3x + 2\tan 3x$

Answer Key

Q1	(A)	Q11	(C)
Q2	(B)	Q12	(A)
Q3	(A)	Q13	(B)
Q4	(B)	Q14	(C)
Q5	(B)	Q15	(B)
Q6	(B)	Q16	(B)
Q7	(C)	Q17	(D)
Q8	(B)	Q18	(A)
Q9	(C)	Q19	(C)
Q10	(B)	Q20	(A)

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